EFFECT OF LONG-ACTING PROGESTAGEN-ONLY INJECTABLE CONTRACEPTIVES ON CARBOHYDRATE METABOLISM AND ITS HORMONAL PROFILE

Kamal Fahmy, Mohammed Abdel-Razik, Mohammed Shaaraway*, Galal Al-Kholy, Samy Saad, Ahmed Wagdi and Mahmoud Al-Azzony**

Departments of Obstetrics and Gynecology, **Clinical Pathology, Benha Faculty of Medicine, and *Endocrinology Unit, Department of Obstetrics and Gynecology, Cairo Faculty of Medicine, Egypt

ABSTRACT

Two groups, each composed of 20 women, who used depo-medroxyprogesterone acetate (DMPA) or norethisterone enanthate (NET-EN) injectable contraceptives were investigated for changes in 75-g OGTT and in the fasting and two-hour post oral glucose load (2-hours) levels of serum insulin, growth hormone, glucagon, cortisol and blood pyruvate. Samples were taken before and 3, 6 and 12 months after use of injectables. DMPA and NET-EN caused significant changes in mean blood glucose and pyruvate and in mean serum insulin, growth hormone and glucagon, but not in mean fasting cortisol. Changes with NET-EN started after 3 months, became maximal after 6 months and reverted to normal after 12 months of use, due to more frequent administration during the first 6 months of use (every 60 $\pm$ 5 days) and to more spacing of the injections (every 84 $\pm$ 5 days) after that. Changes with DMPA started after 3 months, and increased with the duration of use to become maximal after 12 months. Maximal changes were similar with DMPA and NET-EN and consisted of: a significant increase in fasting blood glucose and pyruvate and serum insulin; a significant increase in 2-hour blood glucose and pyruvate, serum insulin, growth hormone and glucagon. Although significant changes in blood glucose levels occurred with both DMPA and NET-EN, yet they did not reach the lower cut-off levels for impaired glucose tolerance in any user. With the same spacing of injections, i.e. every 84 $\pm$ 5 days for NET-EN and every 90 $\pm$ 5 days for DMPA, the effects on various parameters studied were less with NET-EN.

**Correspondence and reprint requests : Prof. Kamal Fahmy, M.D.
127 Al-Merghani Street,
Heliopolis 11361,
Cairo, Egypt.**

Submitted for publication December 19, 1990
Accepted for publication August 19, 1991

INTRODUCTION

Progestagen-only injectable contraceptives, depomedroxy-progesterone acetate (DMPA) and norethisterone enanthate (NET-EN), proved to be successful and popular methods of contraception in many countries. Conflicting results concerning their adverse effects on carbohydrate metabolism were reported. Controversy still exists regarding changes that may occur with the use of these agents in blood glucose and in various hormones involved in carbohydrate metabolism; namely, insulin and the diabetogenic hormones glucagon, growth hormone and cortisol and also in blood pyruvate.

A review of 7 studies on the effects of DMPA on carbohydrate metabolism (1-7), showed that 4 of them (1,2,5,7), revealed significant increase in blood glucose levels, while 3 (3,4,6), revealed no changes. Regarding serum insulin, this showed significant increase at variable times during GTT in 5 studies (1-3,5,6), while 2 (4,7) showed no changes. Four of these studies (1-4) included assay of serum growth hormone, and 3 of them (1,2,4) showed no significant changes, while the fourth (3) showed significant increase in mean serum growth hormone level only at 120 m of GTT. In only one study (6), serum cortisol and pyruvate were assayed, and they showed no significant changes.
Studies that showed changes in carbohydrate metabolism with DMPA, revealed that these changes did not appear with short periods of use, i.e. up to 12 weeks (4), but they appeared after 6 months (1) and increased with longer duration of use (2,3,5,7). Only one author (6), found no significant changes after 12 months. It was, therefore, recommended that DMPA can be used for short-term use for 2 years with a break and that OGTT should be done to check any changes in carbohydrate metabolism (5).

A review of 5 studies on the effects of NET-EN on carbohydrate metabolism (4,6,8-10), revealed that 4 of them (4,6, 8,10), showed no significant changes in blood glucose levels, while one (4) showed an increase after 6 months of NET-EN use, that reverted to normal after 12 months. In 2 of the studies (4,6), serum insulin level was assayed and one of them (4) showed no significant changes, while the other (6), showed a persistant and significant elevation of the pretreatment level. In one study (4), serum growth hormone was assayed and showed no significant changes. In another study (6), serum cortisol and pyruvate were assayed and showed no significant changes. These studies showed that NET-EN did not alter carbohydrate metabolism after 3 months (4,8), after 12 months (6), or even after 26 to 52 months (10). However, changes were reported after 6 months, that reverted to normal after 12 months (9). Therefore, it was recommmended that NET-EN can be used for long-term contraception without adverse effects on carbohydrate metabolism, (6,9,10).

MATERIAL AND METHODS

The material included 40 multiparous women who were selected from those attending the family planning clinic of Benha University Hospital, Egypt. Selection criteria included willingness and fitness for injectables, having at least 2 living children, not lactating and not using any hormonal contraception for at least 6 months. Liver affection was excluded by clinical examination and by normal liver function tests including: serum proteins, bilirubin, alkaline phosphatase, SGPT and SGOT. Bilharziasis was excluded by negative urine and stool analysis for ova and by a negative bilharziasis intradermal immunologic test. Diabetes mellitus was excluded by a normal, standardized 75-g oral glucose tolerance test (OGTT), performed according to the WHO recommendations (11).

Selected women were randomly divided into 2 matched groups, each of 20 women:

1. DMPA group : Received DMPA i.m., in a dose of 150 mg every 90 ± 5 days for 12 months.
2. NET-EN group : Received NET-EN i.m., in a dose of 200 mg every 60 ± 5 days for 6 months, then every 84 ± 5 days for another 6 months.

DMPA and NET-EN dose schedules were the ones recommended by the WHO (12), and used in the Egyptian National Family Planning program.

Before and 3, 6 and 12 months after use of injectables, the following parameters were determined in every woman: A standardized 75-g OGTT (11); fasting serum insulin, growth hormone, glucagon, cortisol and blood pyruvate; two-hour post oral glucose load levels (2-hours) of serum insulin, growth hormone, glucagon and blood pyruvate.

Glucose was determined by the glucose oxidase method, pyruvate by the ultraviolet method of Cozk and Lamprecht (13) and other parameters by radioimmunoassay.

Statistical methods

1. Unpaired student "t" test was used to compare the means between the study groups; DMPA and NET-EN groups.
2. Paired student "t" test was used to compare the means within each group; DMPA and NET-EN groups.

The results were considered significant when the P value was ≤ 0.05.

RESULTS

Before DMPA or NET-EN administration, there were no significant differences in the mean fasting, 30, 60, 90 and

120 serum glucose levels in the OGTT, and in mean age, mean parity and mean weight.

After 12 months of DMPA and NET-EN use, the mean weight with DMPA was 71.2 ± 13 kg compared to a mean pretreatment weight of 69.1 ± 9.8 kg ($P > 0.05$). With NET-EN, the mean weight was 70 ± 11.9 compared to a mean pretreatment weight of 68.9 ± 10.2 kg ($P > 0.05$).

Figures 1 and 2 show the mean serum glucose levels with the 75-g OGTT after 3,6 and 12 months of use, compared to the mean pretreatment levels. With DMPA, there was a significant increase in mean blood glucose at the 60- and 120-minute levels after 3 months, an increase at fasting, 30-minute and 60-minute levels after 6 months, and at all levels after 12 months (Figure 1). With NET-EN, there was a significant increase in mean serum glucose at the fasting, 30-minute and 60-minute levels after 3 months, an increase at all levels after 6 months and the return at all levels to pretreatment values after 12 months (Figure 2).

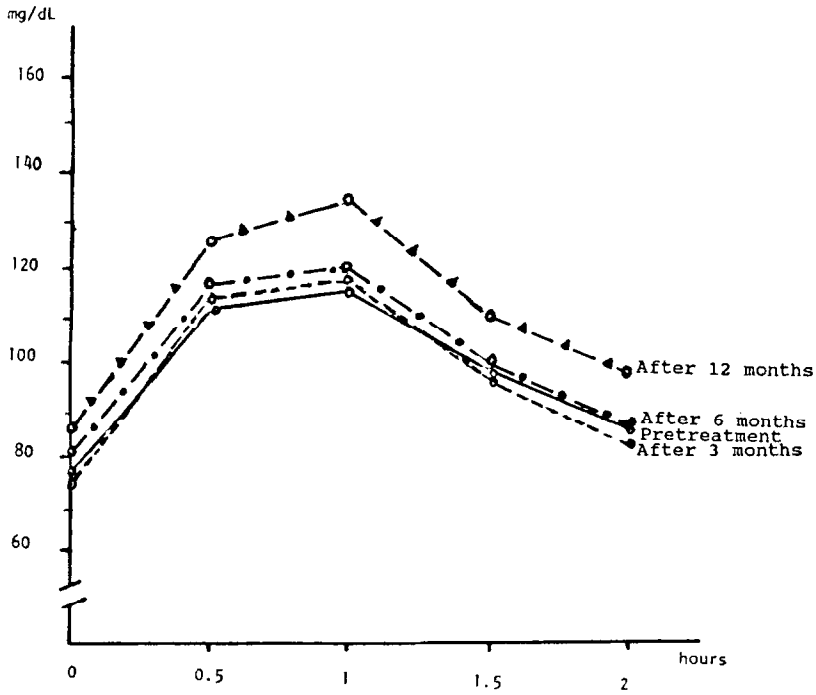

Figure 1: Mean OGTT values with DMPA use.

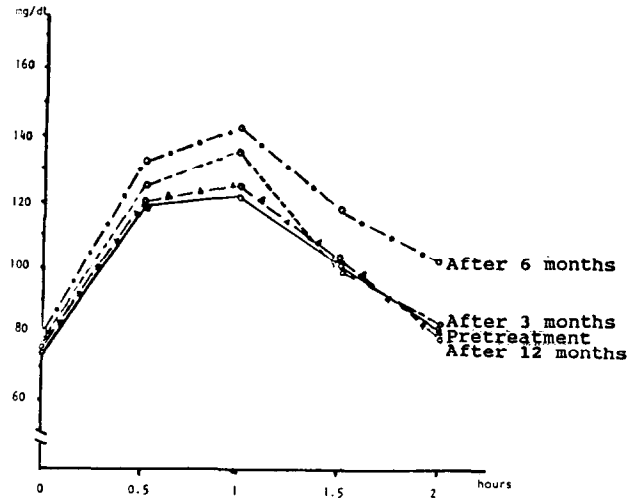

Figure 2: Mean OGTT values with NET-EN use.

Response to glucose load : With the 75-g OGTT, there was a response to glucose load by increase in blood glucose (Figures 1 and 2), in serum insulin (Table I) and in blood pyruvate (Table V), and by decrease in serum growth hormone (Table II) and glucagon (Table III). These responses occurred in the pretreatment assay and also in the assays done at 3, 6 and 12 months after DMPA and NET-EN use.

Comparison of the effect of DMPA and NET-EN use : DMPA use was associated with significantly higher mean 2-hour serum insulin levels after 6 months and mean fasting and 2-hour serum insulin levels after 12 months (Table I), with higher mean 2-hour serum growth hormone and glucagon levels after 12 months (Table II), and with higher mean fasting and 2-hour blood pyruvate levels after 12 months (Table V). On the other hand, NET-EN use was associated with significantly higher mean fasting serum insulin after 6 months (Table I), with higher mean fasting and 2-hour serum glucagon levels after 3 and 6 months (Table III), and with higher mean 2-hour blood pyruvate levels after 3 and 6 months.

The effect of duration of DMPA and NET-EN use : Table I shows that with DMPA there was a significant increase in mean fasting and 2-hour serum insulin after 6 and 12 months, while with NET-EN there was a significant increase in mean fasting serum insulin levels after 3 months, in mean fasting and 2-hour levels after 6 months, and in 2-hour levels after 12 months. Tables II and III show that regarding serum growth hormone and glucagon, there was significant increase

in mean 2-hour levels after 6 and 12 months with DMPA, while with NET-EN this was seen after 6 months only. Table IV concerning serum cortisol, shows no significant changes in mean fasting serum cortisol with both DMPA and NET-EN. Table V shows significant increase in fasting mean blood pyruvate level after 6 months and in both mean fasting and 2-hour blood levels after 12 months with DMPA. NET-EN was associated with significant increase in mean 2-hour blood level after 3 months and in mean fasting and 2-hour levels after 6 months.

Table I. Mean (±SD) serum insulin (nU/ml)

Time / Insulin	Pretreatment Mean	Pretreatment SD	After 3m Mean	After 3m SD	After 6m Mean	After 6m SD	After 12m Mean	After 12m SD
Fasting :								
DMPA (n=20)	7.3	1.5	7.0	1.3	8.7*	1.6	11**	2.2
NET-EN (n=20)	6.9	1.9	6.3	1.9	9.9**	1.7	7.6	2.1
P ++	N.S		N.S		< 0.01		< 0.001	
2-hours :								
DMPA (n=20)	52	19	51	18	129**	44	154**	150
NET-EN (n=20)	47	14	64**	15	108**	32	134**	34
P ++	N.S		N.S		< 0.05		< 0,05	

* P < 0.01 ** P < 0.001 compared to pretreatment mean level (Paired "t" test). P ++ unpaired "t" test.

Table II. Mean (± SD) serum growth hormone (ng/ml)

Time / Growth hormone	Pretreatment Mean	Pretreatment SD	After 3m Mean	After 3m SD	After 6m Mean	After 6m SD	After 12m Mean	After 12m SD
Fasting :								
DMPA (n=20)	2.5	1.1	2.6	1.0	2.4	1.1	2.5	1.1
NET-EN (n=20)	2.8	1.0	2.6	1.3	2.8	1.0	2.8	1.1
P ++	N.S		N.S		N.S		N.S	
2-hours :								
DMPA (n=20)	0.9	0.3	1.1	0.3	1.4+	0.9	1.6*	1.1
NET-EN (n=20)	1.0	0.4	1.1	0.5	1.7**	0.4	1.1	0.5
P ++	N.S		N.S		N.S		< 0.05	

+ P < 0.05 * P < 0.01 ** P < 0.001 compared to pretreatment level (paired "t" test). P ++ unpaired "t" test.

Table III. Mean (± SD) serum glucagon (pg/ml)

Time / Glucagon	Pretreatment Mean	Pretreatment SD	After 3m Mean	After 3m SD	After 6m Mean	After 6m SD	After 12m Mean	After 12m SD
Fasting :								
DMPA (n=20)	362	137	352	87	370	96	379	107
NET-EN (n=20)	433	188	457	166	443	100	387	164
P ++	N.S		< 0.05		< 0.05		N.S	
2-hours :								
DMPA (n=20)	172	107	130	30	247+	80	261+	45
NET-EN (n=20)	201	92	240	168	285*	21	183	77
P ++	N.S		< 0.01		< 0.05		< 0.01	

+ P < 0.05 * P < 0.01 compared to pretreatment mean level (Paired "t" test). P ++ unpaired "t" test.

Table IV. Mean (± SD) fasting serum cortisol (ng/ml)

Time / cortisol	Pretreatment Mean	Pretreatment SD	After 3m Mean	After 3m SD	After 6m Mean	After 6m SD	After 12m Mean	After 12m SD
DMPA (n=20)	119	32	117	30	123	27	127	28
NET-EN (n=20)	120	26	125	30	122	32	128	38
P ++	N.S		N.S		N.S		N.S	

P ++ unpaired "t" test.

Table V. Mean (± SD) blood pyruvate (ng/ml)

Time / Pyruvates	Pretreatment Mean	Pretreatment SD	After 3m Mean	After 3m SD	After 6m Mean	After 6m SD	After 12m Mean	After 12m SD
Fasting :								
DMPA (n=20)	0.50	0.12	0.48	0.10	0.52+	0.10	0.53*	0.12
NET-EN (n=20)	0.48	1.14	0.49	0.13	0.52**	0.13	0.49	0.11
P ++	N.S		N.S.		N.S		<0.05	
2-hours :								
DMPA (n=20)	0.59	0.12	0.61	0.14	6.60	0.13	0.68**	0.16
NET-EN (n=20)	0.57	0.12	0.68**	0.14	0.71**	0.12	0.58	0.13
P ++	N.S		< 0.05		< 0.001		< 0.001	

* P < 0.01 **P < 0.001 compared to pretreatment mean level.
P ++ unpaired "t" test.

Table VI: Summary of changes with DMPA and NET-EN

Parameter	DMPA						NET-EN					
	3mon		6mon		12mon		3mon		6mon		12mon	
	0m	2h	0m	2h	0m	2h	0m	2h	0m	2h	0m	2h
Glucose	-	+	+	-	+	+	+	-	+	+	-	-
Insulin	-	-	+	+	+	+	-	+	+	+	-	+
Growth H.	-	-	-	+	-	+	-	-	-	+	-	-
Glucagon	-	-	-	+	-	+	-	-	-	+	-	-
Cortisol	-	o	-	o	-	o	-	o	-	o	-	o
Pyruvates	-	-	+	-	+	+	-	+	+	+	-	-

+ Significant increase in mean level. - No change 0 not done

DISCUSSION

After injection of 150 mg DMPA, the peak blood level was found to be between 11-27 ng/ml at 4 to 11 days of the injection. These levels fell rapidly to below 10 ng/ml by the 20th day and then more gradually to 3.8-5.5 ng/ml by the 90th day. However, detectable low levels were still present for 6 to 9 months after the injection (14). DMPA in a dose of 150 mg every 12 weeks was associated with a pregnancy rate of 0.0-1.1/100 women-years (15).

After injection of 200 mg NET-EN, the blood level increased rapidly to reach a peak blood level within 5 days. Then the concentration decreased rapidly and was usually undetectable by about 70 days after the injection, although this period may vary from 50 to 120 days (16). With NET-EN, in a dose of 200 mg every 8 to 12 weeks, the pregnancy rate was 0.01-1.3/100 women years (15).

Table VI, which summarizes our findings regarding the effects of DMPA and NET-EN on blood glucose, insulin, growth hormone, cortisol, glucagon and pyruvate shows that the effects of the two drugs were mostly identical, but the time of start and maximal effects were different. The effects of DMPA started after 6 months and became maximal after 12 months, while those with NET-EN started after 3 months and became maximal after 6 months. Changes with NET-EN reverted to pretreatment levels after 12 months, except the 2-hour serum insulin level that remained significantly elevated in spite of reversion of blood glucose and other measured parameters to their pretreatment level. Tuttle and Turkington (3) also found no significant difference between mean

glucose levels at any time during GTT in 18 DMPA users, although the mean serum insulin was abnormally high in 17 of them at fasting, 3 hours or both.

Although significant increase in mean blood glucose levels occurred with both DMPA and NET-EN (Figures 1 and 2), yet the mean 2-hour serum glucose level was 97 ± 19 mg/dL after 12 months of DMPA use and was 102 ± 15mg/dL after 6 months with NET-EN, i.e far below the 2-hour lower cut-off level of impaired glucose level (140 mg/dL). Actually, in not a single DMPA or NET-EN user did the glucose levels reach the lower cut-off levels of impaired glucose tolerance at 30, 60, 90 or 120 minutes. The increase in mean blood glucose level with DMPA and NET-EN occurred first by an increase in peak levels at 30 and 60 minutes, then by lagging of the curve after 90 and 120 minutes.

The earlier, and more significant, increase in mean blood glucose levels seen with NET-EN after 6 months of use may be attributed to the more frequent injections of NET-EN than DMPA (every 60 ± 5 days versus every 90 ± 5 days). After one year of injectables use, significant increase mean blood glucose was manifested with DMPA, but not with NET-EN, where the mean blood glucose levels returned to the pretreatment levels. This may be explained by the extension of dose scheduling of NET-EN to become every 84 ± 5 days instead of every 60 ± 5 days used in the first 6 months, with subsequent less NET-EN effect.

Hormones play an important role in the regulation of plasma glucose concentration. Insulin is the hormone that lowers glucose concentration, while growth hormone, epinephrine, cortisol, glucagon and thyroxine increase it.

The mechanism of producing adverse carbohydrate metabolism with the use of DMPA and NET-EN is not exactly elucidated. A possible mechanism is the glucocorticoid (cortisol-like) action of these drugs (2,3,17). Physiologically, cortisol increases the blood glucose levels, increases insulin secretion, augments action of insulin response to glucose, and raises insulin resistance. Increased release of insulin with DMPA may be a response to extra demand to compensate for the peripheral antagonism of these steroids, and this abnormal response is expected with longer duration of DMPA use, and it reverted to normal as soon as DMPA is neutralized (6). In this study, no significant changes were found in the mean fasting serum cortisol levels with both DMPA and NET-EN, a finding also reported by others (6). This study suggests that the subject's ability to synthesize and release extra quantities of insulin is not impaired by DMPA use. With NET-EN, the study suggests that its use had no such insulin antagonistic effect as DMPA. The return of mean fasting serum insulin values to pretreatment levels after 12 months of NET-EN use does not suggest any

interference with pancreatic insulin synthesis or release, as concomitant glucose load was still exaggerated.

Another possible mechanism is an indirect effect of the cortisol-like action of DMPA and NET-EN through increased production of growth hormone and glucagon, both being diabetogenic. The role of growth hormone in steroid-induced carbohydrate metabolic changes remains unclear. Gershberg et al.(18) reported that fasting growth hormone levels were elevated with DMPA, while Simon et al. (19) reported suppression due to the glucocorticoid action of DMPA. Growth hormone appears to have an antagonizing action on insulin and causes a rise of blood insulin by sensitizing the islets to glucose (20). In our study, suppression of growth hormone levels increased significantly after 6 and 12 months with DMPA and only after 6 months with NET-EN. These results suggest that alteration in serum growth hormone is unlikely the cause of the observed increased fasting serum insulin observed in this study, but it may have a role in the increased insulin response to glucose load.

Glucagon physiologically induces glycolysis, increases gluconeogenesis, stimulates insulin release and stimulates epinephrine from the adrenal medulla which causes rise of fatty acids in the plasma.

Blood pyruvate, a metabolite of glucose metabolism, increased significantly after 6 and 12 months with DMPA and after 3 and 6 months with NET-EN. Increased levels of ketone bodies, free fatty acids and insulin demonstrate increased fatty oxidation. Pyruvate dehydrogenase may be impaired by increased acetyl COA/COA ratio and leads to lowered entry of glycolysis metabolites into the tricarboxylic acid cycle. As a result, pyruvate and glucose levels will increase (18). The concomitant alterations in blood glucose and pyruvate levels shown in our study support this hypothesis.

Changes in serum insulin, growth hormone, glucagon and blood pyruvate occurred more, and persisted longer, with DMPA. The changes in growth hormone, cortisol and glucagon were not the cause of the detected fasting hyperinsulinemia. However, increased growth hormone and cortisol levels might have played a role in increased insulin resistance to glucose load.

REFERENCES

1. **Spellacy WN, McLead AGW, Buhi WV, Birk SA, McCreary BS.** Medroxyprogesterone acetate and carbohydrate metabolism. Measurement of glucose, insulin and growth hormone during 6 months time. Fertil Steril 1970; 21 : 457-63.

2. **Spellacy WN, McLeod AGW, Buhi WC, Birk SA.** The effects of medroxyprogesterone acetate on carbohydrate metabolism. Measurement of glucose, insulin and growth hormone after twelve months use. Fertil Steril 1972; 24 : 239-44.

3. **Tuttle S, Turkington VE.** Effect of medroxyprogestone acetate on carbohydrate metabolism. Obstet Gynecol 1977; 43 : 285-92.

4. **Dhall K, Kumar M, Rastogi G, Devi P.** Short-term effects of norethisterone enanthate and medroxyprogesterone acetate on glucose, insulin, growth hormone and lipids. Fertil Steril 1977; 28 : 156-58.

5. **Liew FM, Ng CSA, Yong YM, Ratnam SS.** Long-term effects of Depo-provera on carbohydrate and lipid metabolism. Contraception 1985; 31 : 51-64.

6. **Amatayakul K, Suriyanon V.** The effect of long-acting injectable contraceptives on carbohydrate metabolism. Int J Gynaecol Obstet 1985; 23 : 361-8.

7. **Liew FM, Charles SA, Heng SH, Ratnam SS.** A comparative study of the metabolic effects of injectable and oral contraceptives. Contraception 1986; 33 : 385-94.

8. **Howard G, Elder MG.** The effect of intramuscular norethisterone enanthate on intravenous glucose tolerance and blood coagulation factors VII and V. Br J Obstet Gynecol 1977; 84 : 618-21.

9. **Mahtab S, Safeya S, Perma K.** Low dose injectable contraceptive norethisterone enanthate 20 mg monthly : metabolic side effects. Contraception 1981; 23 : 23-37.

10. **Howard G, Blair M, Fotherby K, Trayner I, Hamawi A, Elder MG.** Some metabolic effects of long-term use of the injectable contraceptive norethisterone enanthate. Lancet 1982; 1 : 423-25.

11. **WHO.** Expert Committee on Diabetes. Second report. WHO Technical report series, 1980; 646.

12. **WHO.** Multinational comparative clinical evaluation of two long-acting injectable contraceptive steroids : Norethisterone enanthate and medroxyprogestrone acetate. I-Use effectiveness. **Contraception** 1977; 15:513-33.

13. **Cozk R, Lamprecht W.** Pyruvate and phosphenol pyruvate. In : Bergmyet HV, ed. Methods of Enzymatic Analysis. 2nd ed. London : Public Academic Press, 1974 ; 1466.

14. **Cornette JC, Kirton KT, Duhcon CW.** Measurement of medroxyprogestrone acetate by radioimmunoassay. J Clin Endocrinol Metabol 1971; 33 : 459-466.

15. **Wilson ESB.** Injectable contraceptives. In : Loudon N, ed. Handbook of family planning. London : Churchill-Livingstone, 1985; 115.

16. **Sang GW, Fotherbg K, Howard G.** Pharmacokinetics of norethisterone Oenanthate in humans. Contraception 1981; 24:15-27.

17. **Hellman L, Yashida K, Zumoff B, Levin J, Kream J, Fukushima DK.** The effects of medroxyprogesterone acetate on the pitutary adrenal axis. J Clin Endocrinol Metabol 1975; 42-912-17

18. **Gershberg H, Zarilla E, Fernandez A, Hulse M.** Effects of medroxyprogesterone acetate on serum insulin and growth hormone levels in diabetic and potential diabetics. Obstet Gynecol 1969; 33 : 383-9.

19. **Simon S, Schiffer M, Glick SM.** Effect of medroxy-progesterone acetate upon stimulated release of growth hormone. J Clin Enodcinol Metabol 1967; 27 : 1633-36.

20. **Taylor KW.** In : Hormone assays and their clinical applications. Loraine JA and Bell E, eds. 4th ed. London: Churchill-Livingstone, 1976; 330.